

- 2) Canonical weights derived in computing canonical functions are subject to great deal of instability.
- 3) The interpretation of canonical variates is different due to the efforts to maximize the relationship.
- 4) It is difficult to identify meaningful relationship between the subset of independent and dependent variables because precise statistics is yet to be developed.

14.9 LET US SUM UP

Canonical correlation analysis is a useful and powerful technique for exploring the relationships among multiple dependent and independent variables. The technique is primarily descriptive, although it may be used for predictive purposes. In this technique, weighted sums of variables are selected from each of the two sets to form new variables in each of the sets so that the correlation between the new variables in different sets is maximized while the new variables within each set are constrained to be uncorrelated with mean zero and unit variance. Results obtained from a canonical analysis suggest answers to questions concerning the number of ways in which the two sets of multiple variables are related, the strengths of the relationships, and the nature of the relationships defined. Canonical analysis enables the students to combine into a composite measure what otherwise might be an unmanageably large number of bivariate correlations between sets of variables. It is useful for identifying overall relationships between multiple independent and dependent variables, particularly when we have little a priori knowledge about relationships among the data for two sets of variables. Essentially, we can apply canonical correlation analysis to a set of variables that appear to be significantly related.

The CCA is based on two statistical assumptions. First, the correlation coefficient between any two variables is based on a linear relationship. Second, the parent population from which the sample has been drawn is normally distributed. CCA has several advantages and limitations.

14.10 KEY WORDS

Canonical correlation	:	Measure of the strength of the overall relationships between the linear composites (canonical variates) for the independent and dependent variables. In effect, it represents the bivariate correlation between the two canonical variates.
Canonical correlation analysis	:	It is a multivariate statistical analysis that facilitates the study of interrelationships among the sets of multiple dependent variables and multiple independent variables.
Canonical cross-loadings	:	Correlation of each observed independent or dependent variable with the opposite canonical variate. For example, the independent variables are correlated with the dependent canonical variate. They can be interpreted like canonical loadings, but with the opposite canonical variate.
Canonical function	:	Relationship (correlation) between two linear composites (canonical variates). Each canonical function has two canonical variates, one for the set of dependent variables and one for the set of independent variables. The strength of the relationship is given by the canonical correlation.

Canonical loadings	:	Measure of the simple linear correlation between the independent variables and their respective canonical variates. These can be interpreted like factor loadings, and are also known as canonical structure correlations.
Canonical roots	:	Squared canonical correlations, which provide an estimate of the amount of shared variance between the respective optimally weighted canonical variates of dependent and independent variables. It is also known as eigenvalues.
Canonical variates	:	Linear combinations that represent the weighted sum of two or more variables and can be defined for either dependent or independent variables. It is also known as linear composites, linear compounds, and linear combinations.
Multiple regression analysis	:	Multiple regression analysis predicts a single dependent variable from a set of multiple independent variables.

14.11 SOME USEFUL BOOKS

- 1) Alissa Sherry, Robin K. Henson (2005); *Conducting and Interpreting Canonical Correlation Analysis in Personality Research*. Journal of Personality Assessment.
- 2) Hotelling, H. (1936); *Relations between two sets of variables*, Biometrika 28, 321-377
- 3) Johnson, R.A. & Wichern, D.W. (2002); *Applied Multivariate Statistical Analysis*, Pearson Education, Inc.
- 4) Johnson, Dallas E. (1998); *Applied Multivariate Methods for Data Analysis*, International Thomson Publishing Inc.
- 5) Michael S. Lewis-beck Alan Bryman Tim Futing Liao (2004); *The sage encyclopedia of Social Sciences Research Methods* vol. 1 page no. 83
- 6) Magnus Borga; Canonical Correlation. A tutotial www.cs.cmu.edu/~tom/10701-spll/sides/cca-tutorial.pdf
- 7) S. Press James (2005); *Applied Multivariate Analysis*. Dover publication.inc.

14.12 ANSWERS OR HINTS TO CHECK YOUR PROGRESS

Check Your Progress I

- 1) See Section 14.1
- 2) See Section 14.6
- 3) See Section 14.6

14.13 EXERCISES

- 1) Under what circumstances would you select canonical correlation analysis instead of multiple regressions as the appropriate statistical technique?
- 2) Discuss in details the procedure for computation of the canonical correlation.
- 3) What are the limitations associated with canonical correlation analysis?

UNIT 15 CLUSTER ANALYSIS

Structure

- 15.0 Objectives
- 15.1 Introduction
- 15.2 Cluster Analysis: Concept and Meaning
- 15.3 Steps and Algorithm Involved in Cluster Analysis
- 15.4 Methods of Cluster Analysis
- 15.5 Partitioning Cluster Methods
- 15.6 Hierarchical Cluster Methods
- 15.7 Other Approaches: Two-step Cluster Analysis
- 15.8 Interpretation of the Results
- 15.9 Let Us Sum Up
- 15.10 Key Words
- 15.11 Some Useful Books
- 15.12 Answers or Hints to Check Your Progress Exercises
- 15.13 Exercises

15.0 OBJECTIVES

After going through this unit, you will be able to:

- state the concept and purpose of cluster analysis;
- list the steps to be followed in cluster analysis;
- explain the different approaches to cluster analysis; and
- to learn how to apply cluster analysis in analyzing economic problems and interpret its results.

15.1 INTRODUCTION

In social sciences, data set usually take the form of observations on unit of analysis for a set of variables. You as a student of economics may be interested to identify groups of similar objects like countries, enterprises, households on the basis of selected variables like unemployment rate of men and women in different countries, deprivation indicators of households etc. Similarly you may need to segment customers into a small number of groups for additional analysis and marketing activities. Further researcher may aim to find out spatial or temporal pattern in human developments, educational development etc. In such cases, a simple classification of units into sub-groups is required for the purpose of analysis because classes or conceptually meaningful groups (clusters) share common characteristics and play an important role in analysis and description of the world. Cluster analysis is a useful technique in such situations for data analysis. For example, cities can be grouped or clustered in

terms of their social, economic and demographic characteristics. Similarly, people can be clustered in terms of their psychological profiles or other attributes they possess. Hence, in this unit, we shall discuss various clustering methods and algorithm involved in cluster analysis. Let us begin with explaining the concept of cluster analysis.

15.2 CLUSTER ANALYSIS: CONCEPT AND MEANING

Classes or conceptually meaningful groups of objects sharing common characteristics, play an important role in how people analyze and describe the world. The creation of groups of similar objects or variables based upon measured characteristics is an important method of analysis which we call cluster analysis. This is an important method of multi-variate analysis. After collecting the data, we divide the observations into groups (clustering) and assign the labels to these groups. Thus cluster analysis is the study of techniques for automatically finding classes. Cluster analysis is a strong tool of the multivariate exploratory data analysis. It involves a great amount of techniques, methods and algorithms which can be applied in various fields of social sciences including economics.

Let us remember that Grouping, or clustering, is distinct from the classification methods. Classification pertains to known number of groups with the purpose to assign new observations to one of these groups. Cluster analysis is a more primitive technique in the sense that no assumptions are made concerning the number of groups or the group structure. Grouping is done on the basis of similarities or distances (dissimilarities). The inputs required are similarity measures or data from which similarities can be computed. The general practical application of cluster analysis is whether the investigator knows enough about the problem to distinguish "good" groupings from "bad" groupings. This can be done by enumerating all the possible groupings and select the "best" ones for further study.

Cluster analysis aims to discover natural groupings of the items (or variables). For this, we need to first develop a quantitative scale to measure the association (similarity) between objects. Since clustering involves a collection of data objects which are similar to one another within the cluster and dissimilar to objects in other clusters, a cluster of data objects can be treated as an implicit class. Due to this it is also sometimes referred to as 'automatic classification'. In some applications, it is also known as 'data segmentation' because clustering divides large data sets into groups based on their similarities. Cluster analysis enables us to detect outliers. As a branch of statistics, cluster analysis has been extensively studied with the main focus on distance-based cluster analysis and has been built into statistical analysis software packages or systems such as SPSS, SAS and S-Plus. An illustration of cluster analysis using SPSS shall be discussed towards the end of the unit. It is, however, first important to understand the steps and algorithm involved in conducting cluster analysis and the main methods used under this analytic technique in research.

15.3 STEPS AND ALGORITHM INVOLVED IN CLUSTER ANALYSIS

The following five broad steps are involved in conducting cluster analysis:

- 1) Measuring the relevant variables (both quantitative and categorical) that can be included,
- 2) Creating a (dis) similarity Matrix for an appropriate measure of (dis) similarity,

Distances are used to measure (dis) similarity between the samples of two variables. To determine the extend of (dis) similarity we construct what is known as (dis) similarity matrix. Every entry of (dis) similarity matrix determine using the notion of distance, more formally metric. Various sort of metric can be employed each being suitable to suit the particular sort of data set.

The metric that is usually employed is known as *minskowi* metric, the formula of which is as under:

$$d(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^k \right)^{\frac{1}{k}}$$

where $x_i = (x_1, x_2, x_3, \dots, x_n)$ and $y_i = (y_1, y_2, y_3, \dots, y_n)$ are two n – dimensional data objects and k is a positive integer.

For $k = 1$ and $k = 2$ the *Minskowi* metric becomes equal to *Manhattan* and *Euclidean* metric respectively.

- 3) Creating one or more clustering's via a clustering algorithm. An example for clustering algorithm:

For the sake of convenience, we shall consider a sub matrix from the table of (dis) similarity matrix shown in table 1.

Let us take

$$A = \begin{pmatrix} 0 & & & & & \\ 1 & 0 & & & & \\ 0 & 1 & 0 & & & \\ 1 & 2 & 1 & 0 & & \\ 0 & 1 & 0 & 1 & 0 & \\ 16 & 17 & 16 & 9 & 16 & 0 \end{pmatrix}$$

Now, we implement the algorithm.

Step 1: The entries in A are squared Euclidean distances. We first take their positive square root and call the new matrix as D_1

$$D_1 = \begin{pmatrix} 1 & 0 & & & & \\ 2 & 1 & 0 & & & \\ 3 & 0 & 1 & 0 & & \\ 4 & 1 & \sqrt{2} & 1 & 0 & \\ 5 & 0 & 1 & 0 & 1 & 0 \\ 6 & 4 & \sqrt{17} & 4 & 3 & 4 & 0 \end{pmatrix}$$

We label rows as 1, 2, 3, 4, 5, 6. Taking the row 1 as a fixed variable, we now locate the variable closest to it. We observe that there are two such choices (1,3) & (1,5) as both have distance zero. We take (1,3) and put 1 & 3 in the same cluster. Now we calculate the distance of (1,3) from remaining rows with the understanding that

$$d_{ij} = |d_{ij} - d_{ji}| \quad \forall i, j = 1(1)5$$

$$d_{(13)2} = \min\{d_{12}, d_{32}\} = \min\{1, 1\} = 1$$

$$d_{(13)4} = \min\{d_{14}, d_{34}\} = \min\{1, 1\} = 1$$

$$d_{(13)5} = \min\{d_{15}, d_{35}\} = \min\{0, 0\} = 0$$

$$d_{(13)6} = \min\{d_{16}, d_{36}\} = \min\{4, 4\} = 4$$

Hence, now D_2 becomes

$$D_2 = \begin{matrix} & \begin{matrix} 13 \\ 2 \\ 4 \\ 5 \\ 6 \end{matrix} \end{matrix} \begin{pmatrix} 0 & & & & \\ 1 & 0 & & & \\ 1 & \sqrt{2} & 0 & & \\ 0 & 1 & 1 & 0 & \\ 4 & \sqrt{17} & 3 & 4 & 0 \end{pmatrix}$$

We observe that because 1 & 3 were clustered together, therefore, 3rd row so does the 3rd column from D_1 gets deleted.

Step 2: We now see that taking (13) as the fixed variable, the variable closest to (13) in the matrix D_2 is row 5 as their distance is zero. Hence, we cluster together (13) & 5.

Now we calculate distance of (13)5 from the remaining variable to construct matrix D_3 .

Hence now D_3 becomes

$$D_3 = \begin{matrix} & \begin{matrix} (13)5 \\ 2 \\ 4 \\ 6 \end{matrix} \end{matrix} \begin{pmatrix} 0 & & & \\ 1 & 0 & & \\ 1 & \sqrt{2} & 0 & \\ \sqrt{17}-1 & \sqrt{17} & 3 & 0 \end{pmatrix}$$

We can continue like this and it is entirely up to us when to stop depending on how many clusters we want. This method of creating clusters is applicable in both hierarchical as well as non hierarchical approach to cluster analysis.

- 4) providing some assessment of the obtained clustering(s), and
- 5) Interpreting the clustering(s) in substantive terms.

Assessment and interpretation of the obtained clustering (s) has been explained in section 15.8 with the help of an illustration.

15.4 METHODS OF CLUSTER ANALYSIS

Imagine you are going to undertake fieldwork in Uttar Pradesh next week and for doing so you wish to divide your group into 5 teams and assign 10 villages under one team each. Strategically, you would like to identify those set of 10 villages to bear somewhat similar characteristics as much possible. Suppose that the district information is also unavailable leaving manual grouping difficult. You need a clustering tool to help you in this scenario. Clustering is a process of grouping a set of objects into multiple groups or clusters so that objects having high similarity are in same cluster but having dissimilarity in other clusters. Similarities and dissimilarities are generally assessed based on attribute values describing objects and involve distance measures.

Creation of simple group structure from a complex data set require a measure of “closeness” or “similarity”. Often a great deal of subjectivity is involved in the choice of a similarity measure. Important considerations include the nature of the variables (discrete, continuous, binary), scales of measurement (nominal, ordinal, interval, ratio), and subject matter knowledge. When items (units or cases) are clustered, proximity is usually indicated by some sort of distance. On the other hand, variables are usually grouped on the basis of correlation coefficients or like measures of association.

An advantage of using Cluster analysis is to represent data in graphical way as it offers several possibilities to do that. Dendrogram or other graphical solutions can be used to visually examine linkages between observations. A dendrogram is a tree-like structure commonly used to represent the process of hierarchical clustering. It shows how objects are grouped together (in an agglomerative method) or partitioned (in a divisive method) on a step-by-step basis.

Two types of clustering algorithms are generally reported, namely non-hierarchical and hierarchical. Partitioning is the most extensively used non-hierarchical method and would be discussed in detail in this unit. While partitioning method classifies objects into a specified number of groups (say k), hierarchical method does not construct a single partition with k clusters, but deals with all values of k in the interval $[1, n]$ where n is the total number of objects. Other types of non-hierarchical clustering methods are density-based and grid-based which are beyond the scope of this unit. Non-hierarchical cluster analysis tends to be used when large data sets are involved. It is sometimes preferred because it allows subjects to move from one cluster to another (this is not possible in hierarchical cluster analysis where a subject, once assigned, cannot move to a different cluster). The two major disadvantages of non-hierarchical cluster analysis are:

- 1) It is often difficult to know how many clusters you are likely to have and therefore the analysis may have to be repeated several times, and
- 2) It can be very sensitive to the choice of initial cluster centres.

15.5 PARTITIONING CLUSTERING METHODS

As introduced earlier, a partitioning method classifies objects into say, k groups which together must satisfy the following criteria:

Each group must contain at least one object and each object must belong to one group. To put it simply, it implies that k should be less than or equal to n , where n is the total number of objects. In other words, partitioning methods conduct one-level partitioning on data sets. The basic partitioning methods typically adopt 'exclusive cluster separation' i.e. each object must belong to exactly one group. Stratified random sampling lend an interesting example of partitioning methods. Suppose a household sample size of 100 has to be collected from Delhi representing all five geographical zones of Delhi i.e. East, West, North, South and Central zones. In this case the five zones are the 5 groups and are pre-specified in number. Each zone must contain at least one household and each household in the sample must belong to one zone. Clearly in this example, the number of groups k (here 5) are less than the number of objects n (here 100).

In general, after initial partitioning, the 'iterative relocation technique' attempts to improve the partitioning by moving objects from one group to another. When it comes to judging a good partitioning, it can be inferred on the basis of whether objects in the same cluster are 'close' or related to each other whereas objects in different clusters are 'far apart'.

We shall now discuss one of the most frequently used partitioning method in social sciences and in statistical analysis and that is the **K-means method**. It is a Centroid based technique. K-means clustering is used to split the observations into K clusters/groups and test what are the main characteristics of these clusters/groups. To understand it in simple words, let us assume that we have a data set D which contains n objects in Euclidian space i.e. two dimensional space. Partitioning methods first distribute the objects in data set D into k clusters such as $X_1, X_2, \dots, X_k$, wherein each X_i is an element of D and $X_i \cap X_j = \emptyset$. Thereafter an objective function is defined to assess the partitioning quality so that within a cluster there are similar objects while other clusters have dissimilar objects.

As mentioned earlier, K-means is a centroid based method which uses the centroid of a cluster to represent that cluster. Conceptually, centroid is nothing but its center point or mean. It is, many a times, defined as the mean of the objects assigned to the cluster. After calculating the mean, this methods measures the distance of objects from the centroid in the Euclidian space. The quality of a cluster is measured by the within-cluster variation, which is the sum of squared error between all objects in X_i and the centroid c_i .

How does the k-means algorithm work? Han, Kember and Pei have explained the workings in a very simple manner. The k-means algorithm defines the centroid of a cluster as the mean value of the points within the cluster.

- a) First, it randomly selects k objects in D , each of which initially represent a cluster mean or center.
- b) For each of the remaining objects, an object is assigned to the cluster to which it is most similar, based on the Euclidian distance between the object and the cluster mean.
- c) K-means algorithm thereafter improves the intra-cluster variation iteratively.
- d) For each cluster, it computes the new mean using the objects assigned to the cluster in the previous iteration.
- e) All objects are then reassigned using updated means as the new cluster centers.

- f) Iterations continue until the assignment is stable i.e. clusters formed in the current step are equal to the clusters formed in previous step.

Check Your Progress 1

- 1) Explain the concept and intuition of Cluster analysis. What are the steps involved in carrying out cluster analysis?

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- 2) Which are the broad methods used in cluster-analysis? Explain the k-means method clearly indicating the steps involved in it.

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15.6 HIERARCHICAL CLUSTERING METHODS

In some situations, you may like to partition data into groups at different levels as in a hierarchy. A Hierarchical clustering method works by grouping data objects into a hierarchy or 'tree' of clusters. By doing so it becomes easy to summarize data. This will enable you to visually interpret it. For example, the employees of a college may be divided on the basis of their nature of appointment: Permanent or Temporary and one may further partition them into smaller sub-groups like Professor, Associate Professor, Reader, Assistant Professor, etc. All these groups form a hierarchy. You can easily characterize or summarize the data in a hierarchy and can use it to find the average working experience or salaries of Professors and of Assistant Professors.

You may use hierarchical clustering methods to analyze the hierarchical structure in an organization or dataset or may also use to discover an underlying hierarchical structure. In the study of evolution, hierarchical clustering has been used to group animals according to their biological features which has enabled scientists to discover evolutionary paths, which are a hierarchy of species.

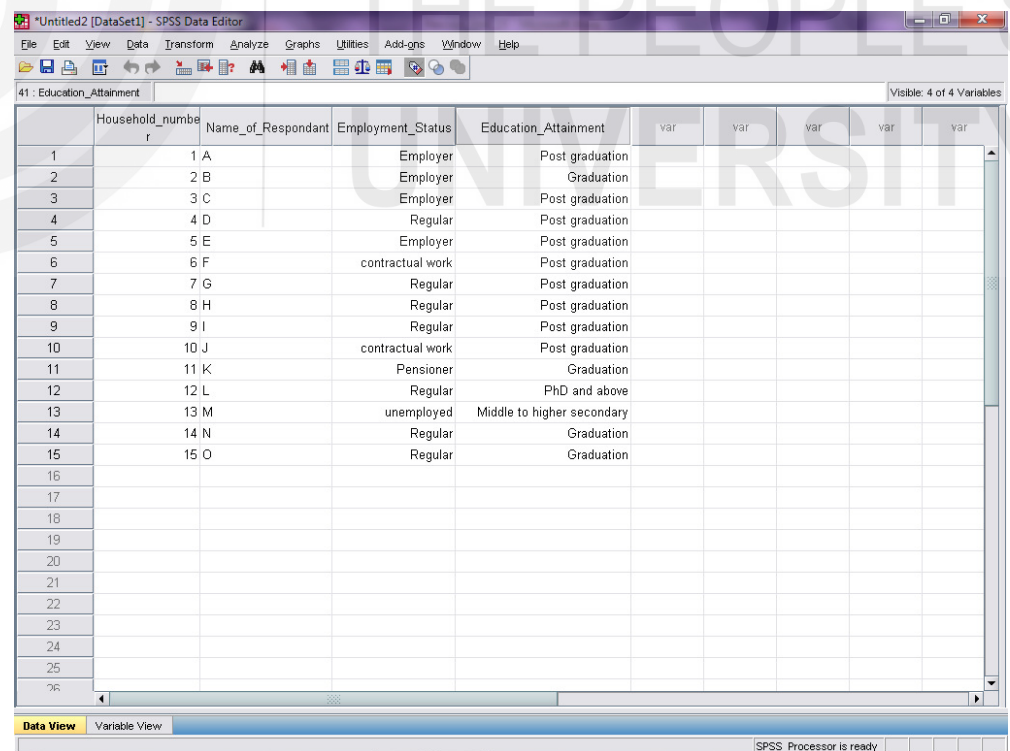
Hierarchical algorithms do not construct a single partition with k clusters, but they deal with all values of k in the interval $[1, n]$. Hierarchical clustering methods are of two types: Agglomerative and Divisive. Agglomerative hierarchical methods start with each unit in its own cluster and systematically merges units and clusters until all units form a single cluster. Thus, there are initially as many clusters as objects. The most similar objects are first grouped, and these initial groups are merged according to their similarities. Eventually, as the similarity decreases, all subgroups are fused into a single cluster. An agglomerative hierarchical method uses a bottom-up strategy. **Han, Kamber**

and **Pei** explain that it typically starts by letting each object form its own cluster and iteratively merges clusters into larger and larger clusters, until all objects are in a single cluster. The single cluster becomes the hierarchy's root. For merging, it finds two clusters closest to each other and combines to form one cluster. It requires at most n iterations as two clusters are merged per iteration and each cluster contains at least one object.

Divisive hierarchical methods work in the reverse direction. Initially all objects belong to the same cluster; then, iteratively, a cluster is chosen according to a selection criterion and bi-partitioned such that the objects in one subgroup are "far from" the objects in the other. These subgroups are then further divided into dissimilar subgroups; the process continues until there are as many clusters as objects that is, until each object forms a group. Thus it employs a top-down approach. It starts by placing all objects in one cluster, which is the hierarchy's root. It then divides the root cluster into several smaller sub-clusters and repeatedly partitions them into smaller clusters.

In other words, hierarchical clustering techniques proceed by either a series of successive mergers or a series of successive divisions. In either agglomerative or divisive hierarchical clustering, a user can specify the desired number of clusters as a termination condition. The results of both agglomerative and divisive methods may be displayed in the form of a two-dimensional diagram known as dendrogram. However, this graph is generally useful for relatively small data sets only. In case of large data sets or sample size, individual objects cannot be identified. As we shall see, the dendrogram illustrates the mergers or divisions that have been made at successive levels.

Illustration: Suppose we take a sample of size 15 and wish to examine the inter-relationship of employment status and educational attainment as shown in Figure 15.1. Cluster Analysis helps in clustering objects which are similar in attributes.



	Household_number	Name_of_Respondant	Employment_Status	Education_Attainment	var	var	var	var	var
1		1 A	Employer	Post graduation					
2		2 B	Employer	Graduation					
3		3 C	Employer	Post graduation					
4		4 D	Regular	Post graduation					
5		5 E	Employer	Post graduation					
6		6 F	contractual work	Post graduation					
7		7 G	Regular	Post graduation					
8		8 H	Regular	Post graduation					
9		9 I	Regular	Post graduation					
10		10 J	contractual work	Post graduation					
11		11 K	Pensioner	Graduation					
12		12 L	Regular	PhD and above					
13		13 M	unemployed	Middle to higher secondary					
14		14 N	Regular	Graduation					
15		15 O	Regular	Graduation					
16									
17									
18									
19									
20									
21									
22									
23									
24									
25									

Fig. 15.1: Sample data for Cluster Analysis

Source: Assessment of Human wellbeing: A Multidimensional Approach unpublished study undertaken by prof. Narayan Prasad